

Babu Banarasi Das University



NO SQL Assignment

SUBMITTED TO

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ACKNOWLEDGEMENT

I sincerely express my gratitude to my teacher for giving me the opportunity to work on this NoSQL project titled "Product Inventory Management System Using MongoDB".

This project helped me understand the basic concepts of NoSQL databases and learn how MongoDB is used to create, retrieve, update, sort and delete data. I also learned how different query operators such as greater than, less than, equal to, AND, and OR are used in MongoDB.

I am thankful to my teacher for the guidance and support provided throughout the completion of this project.

INTRODUCTION

MongoDB is a popular NoSQL database management system that stores data in the form of documents. Unlike traditional relational databases, MongoDB does not require data to be stored in fixed rows and columns.

In this project, a Product Inventory Management System has been created using MongoDB. The database contains information about 50 products, including their product ID, name, category, price, and warehouse location.

Different MongoDB operations have been performed on the product data, such as inserting, displaying, updating, deleting, sorting, counting, and searching records. Comparison operators such as \$eq, \$gt, \$lt, \$gte, and \$lte have also been demonstrated.

OBJECTIVES

- To understand the concept of NoSQL databases.

- To learn the basic features of MongoDB.

- To create a database and collection.

- To insert 50 product records.

- To perform CRUD operations.

- To use comparison operators.

- To perform sorting and filtering.

- To understand AND and OR operations.

1. CREATEDATABASE

MongoDB Query:

use Inventory

Explanation: Creates or switches to the Inventory database.

```
test> use Inventory
switched to db Inventory
Inventory> █
```

2. CREATECOLLECTION

MongoDB Query:

db.createCollection("products")

Explanation: Creates a collection named products.

```
Inventory> db.createCollection("products")
{ ok: 1 }
Inventory> █
```

3. INSERT 50 PRODUCTS

MongoDB Query:

db.products.insertMany([])

```
{ productId: 1, name: "Wireless Mouse", category: "Electronics", price: 880, warehouse:
"Lucknow" },
{ productId: 2, name: "Basmati Rice 5kg", category: "Grocery", price: 760, warehouse:
"Jaunpur" },
{ productId: 3, name: "Bluetooth Speaker", category: "Electronics", price: 920,
warehouse: "Kanpur" },
{ productId: 4, name: "Sunflower Oil 1L", category: "Grocery", price: 690, warehouse:
"Ayodhya" },
{ productId: 5, name: "LED Monitor", category: "Electronics", price: 810, warehouse:
"Varanasi" },
{ productId: 6, name: "Office Chair", category: "Furniture", price: 850, warehouse:
"Lucknow" },
{ productId: 7, name: "Smartphone Charger", category: "Electronics", price: 730,
warehouse: "Prayagraj" },
{ productId: 8, name: "Laptop Stand", category: "Electronics", price: 900, warehouse:
"Lucknow" },
{ productId: 9, name: "Wooden Bookshelf", category: "Furniture", price: 780, warehouse:
"Gorakhpur" },
{ productId: 10, name: "Wheat Flour 5kg", category: "Grocery", price: 650, warehouse:
"Kanpur" },
```

{ productId: 11, name: "Study Table", category: "Furniture", price: 710, warehouse: "Jaunpur" },
{ productId: 12, name: "USB Hub", category: "Electronics", price: 950, warehouse: "Lucknow" },
{ productId: 13, name: "Green Tea Pack", category: "Grocery", price: 840, warehouse: "Ayodhya" },
{ productId: 14, name: "Smartwatch", category: "Electronics", price: 870, warehouse: "Prayagraj" },
{ productId: 15, name: "Dining Table", category: "Furniture", price: 740, warehouse: "Varanasi" },
{ productId: 16, name: "Organic Honey", category: "Grocery", price: 890, warehouse: "Lucknow" },
{ productId: 17, name: "Power Bank", category: "Electronics", price: 820, warehouse: "Kanpur" },
{ productId: 18, name: "Masala Tea", category: "Grocery", price: 580, warehouse: "Gonda" },
{ productId: 19, name: "Sofa Set", category: "Furniture", price: 910, warehouse: "Lucknow" },
{ productId: 20, name: "Wireless Earbuds", category: "Electronics", price: 630, warehouse: "Bareilly" },
{ productId: 21, name: "Peanut Butter Jar", category: "Grocery", price: 770, warehouse: "Lucknow" },
{ productId: 22, name: "Webcam", category: "Electronics", price: 860, warehouse: "Jaunpur" },
{ productId: 23, name: "Bed Frame", category: "Furniture", price: 720, warehouse: "Kanpur" },
{ productId: 24, name: "WiFi Router", category: "Electronics", price: 800, warehouse: "Ayodhya" },
{ productId: 25, name: "Cornflakes Box", category: "Grocery", price: 670, warehouse: "Prayagraj" },
{ productId: 26, name: "Wardrobe", category: "Furniture", price: 940, warehouse: "Lucknow" },
{ productId: 27, name: "Mechanical Keyboard", category: "Electronics", price: 880, warehouse: "Varanasi" },
{ productId: 28, name: "Almonds Pack", category: "Grocery", price: 610, warehouse: "Gorakhpur" },
{ productId: 29, name: "Tablet Cover", category: "Electronics", price: 750, warehouse: "Kanpur" },
{ productId: 30, name: "Coffee Table", category: "Furniture", price: 790, warehouse: "Lucknow" },
{ productId: 31, name: "Pasta Pack", category: "Grocery", price: 680, warehouse: "Jaunpur" },
{ productId: 32, name: "Extension Board", category: "Electronics", price: 900, warehouse: "Lucknow" },

```
{ productId: 33, name: "Recliner Chair", category: "Furniture", price: 550, warehouse:
"Ayodhya" },
{ productId: 34, name: "Olive Oil", category: "Grocery", price: 830, warehouse: "Kanpur" },
{ productId: 35, name: "Cooling Fan", category: "Electronics", price: 960, warehouse:
"Prayagraj" },
{ productId: 36, name: "Shoe Rack", category: "Furniture", price: 700, warehouse:
"Lucknow" },
{ productId: 37, name: "Coffee Beans", category: "Grocery", price: 810, warehouse:
"Varanasi" },
{ productId: 38, name: "Pen Drive", category: "Electronics", price: 660, warehouse:
"Gorakhpur" },
{ productId: 39, name: "TV Unit", category: "Furniture", price: 930, warehouse:
"Lucknow" },
{ productId: 40, name: "Graphics Card", category: "Electronics", price: 850, warehouse:
"Kanpur" },
{ productId: 41, name: "Mixed Spices Pack", category: "Grocery", price: 740, warehouse:
"Jaunpur" },
{ productId: 42, name: "Dressing Table", category: "Furniture", price: 890, warehouse:
"Lucknow" },
{ productId: 43, name: "Smart Bulb", category: "Electronics", price: 620, warehouse:
"Ayodhya" },
{ productId: 44, name: "Dry Fruits Box", category: "Grocery", price: 780, warehouse:
"Prayagraj" },
{ productId: 45, name: "Bunk Bed", category: "Furniture", price: 840, warehouse:
"Lucknow" },
{ productId: 46, name: "HDMI Cable", category: "Electronics", price: 730, warehouse:
"Kanpur" },
{ productId: 47, name: "Sugar 1kg", category: "Grocery", price: 590, warehouse: "Gonda" },
{ productId: 48, name: "Filing Cabinet", category: "Furniture", price: 870, warehouse:
"Lucknow" },
{ productId: 49, name: "Portable SSD", category: "Electronics", price: 640, warehouse:
"Bareilly" },
{ productId: 50, name: "Salt 1kg", category: "Grocery", price: 910, warehouse: "Varanasi" },
```

```
'19': ObjectId('6ab7df10fa70e9ba4d075e0a'),
'20': ObjectId('6ab7df10fa70e9ba4d075e0b'),
'21': ObjectId('6ab7df10fa70e9ba4d075e0c'),
'22': ObjectId('6ab7df10fa70e9ba4d075e0d'),
'23': ObjectId('6ab7df10fa70e9ba4d075e0e'),
'24': ObjectId('6ab7df10fa70e9ba4d075e0f'),
'25': ObjectId('6ab7df10fa70e9ba4d075e10'),
'26': ObjectId('6ab7df10fa70e9ba4d075e11'),
'27': ObjectId('6ab7df10fa70e9ba4d075e12'),
'28': ObjectId('6ab7df10fa70e9ba4d075e13'),
'29': ObjectId('6ab7df10fa70e9ba4d075e14'),
'30': ObjectId('6ab7df10fa70e9ba4d075e15'),
'31': ObjectId('6ab7df10fa70e9ba4d075e16'),
'32': ObjectId('6ab7df10fa70e9ba4d075e17'),
'33': ObjectId('6ab7df10fa70e9ba4d075e18'),
'34': ObjectId('6ab7df10fa70e9ba4d075e19'),
'35': ObjectId('6ab7df10fa70e9ba4d075e1a'),
'36': ObjectId('6ab7df10fa70e9ba4d075e1b'),
'37': ObjectId('6ab7df10fa70e9ba4d075e1c'),
'38': ObjectId('6ab7df10fa70e9ba4d075e1d'),
'39': ObjectId('6ab7df10fa70e9ba4d075e1e'),
'40': ObjectId('6ab7df10fa70e9ba4d075e1f'),
'41': ObjectId('6ab7df10fa70e9ba4d075e20'),
'42': ObjectId('6ab7df10fa70e9ba4d075e21'),
'43': ObjectId('6ab7df10fa70e9ba4d075e22'),
'44': ObjectId('6ab7df10fa70e9ba4d075e23'),
'45': ObjectId('6ab7df10fa70e9ba4d075e24'),
'46': ObjectId('6ab7df10fa70e9ba4d075e25'),
'47': ObjectId('6ab7df10fa70e9ba4d075e26'),
'48': ObjectId('6ab7df10fa70e9ba4d075e27'),
'49': ObjectId('6ab7df10fa70e9ba4d075e28')
}
Inventory> █
```

Explanation: Inserts multiple product documents into the collection.

4. DISPLAY ALL PRODUCTS

MongoDB Query:

```
db.products.find()
```

Explanation: Retrieves documents from the products collection.

```
Inventory> db.products.find()
{
  _id: ObjectId('6ab7df10fa70e9ba4d075df7'),
  productId: 1,
  name: 'Wireless Mouse',
  category: 'Electronics',
  price: 880,
  warehouse: 'Lucknow'
}
{
  _id: ObjectId('6ab7df10fa70e9ba4d075df8'),
  productId: 2,
  name: 'Basmati Rice 5kg',
  category: 'Grocery',
  price: 760,
  warehouse: 'Jaunpur'
}
{
  _id: ObjectId('6ab7df10fa70e9ba4d075df9'),
  productId: 3,
  name: 'Bluetooth Speaker',
  category: 'Electronics',
  price: 920,
  warehouse: 'Kanpur'
}
{
  _id: ObjectId('6ab7df10fa70e9ba4d075dfa'),
  productId: 4,
  name: 'Sunflower Oil 1L',
  category: 'Grocery',
  price: 690,
  warehouse: 'Ayodhya'
}
{
```

5. EQUAL TO —\$eq

MongoDB Query:

```
db.products.find({ price: { $eq: 880 } })
```

Explanation: Finds products whose price is exactly 880.

```
Inventory> db.products.find({ price: { $eq: 880 } })
{
  _id: ObjectId('6ab7df10fa70e9ba4d075df7'),
  productId: 1,
  name: 'Wireless Mouse',
  category: 'Electronics',
  price: 880,
  warehouse: 'Lucknow'
}
{
  _id: ObjectId('6ab7df10fa70e9ba4d075e11'),
  productId: 27,
  name: 'Mechanical Keyboard',
  category: 'Electronics',
  price: 880,
  warehouse: 'Varanasi'
}
Inventory> █
```

6. GREATER THAN—\$gt

MongoDB Query:

```
db.products.find({ price: { $gt: 850 } })
```

Explanation: Finds products whose price is greater than 850.

```
Inventory> db.products.find({ price: { $gt: 850 } })
{
  _id: ObjectId('6ab7df10fa70e9ba4d075df7'),
  productId: 1,
  name: 'Wireless Mouse',
  category: 'Electronics',
  price: 880,
  warehouse: 'Lucknow'
}
{
  _id: ObjectId('6ab7df10fa70e9ba4d075df9'),
  productId: 3,
  name: 'Bluetooth Speaker',
  category: 'Electronics',
  price: 920,
  warehouse: 'Kanpur'
}
{
  _id: ObjectId('6ab7df10fa70e9ba4d075dfe'),
  productId: 8,
  name: 'Laptop Stand',
  category: 'Electronics',
  price: 900,
  warehouse: 'Lucknow'
}
{
  _id: ObjectId('6ab7df10fa70e9ba4d075e02'),
  productId: 12,
  name: 'USB Hub',
  category: 'Electronics',
  price: 950,
  warehouse: 'Lucknow'
}
{
```

7. LESS THAN—\$lt

MongoDB Query:

```
db.products.find({ price: { $lt: 600 } })
```

Explanation: Finds products whose price is less than 600.

```
Inventory> db.products.find({ price: { $lt: 600 } })
{
  _id: ObjectId('6ab7df10fa70e9ba4d075e08'),
  productId: 18,
  name: 'Masala Tea',
  category: 'Grocery',
  price: 580,
  warehouse: 'Gonda'
}
{
  _id: ObjectId('6ab7df10fa70e9ba4d075e17'),
  productId: 33,
  name: 'Recliner Chair',
  category: 'Furniture',
  price: 550,
  warehouse: 'Ayodhya'
}
{
  _id: ObjectId('6ab7df10fa70e9ba4d075e25'),
  productId: 47,
  name: 'Sugar 1kg',
  category: 'Grocery',
  price: 590,
  warehouse: 'Gonda'
}
Inventory> █
```

8. GREATER THAN OR EQUAL TO — \$gte

MongoDB Query:

```
db.products.find({ price: { $gte: 900 } })
```

Explanation: Finds products whose price is 900 or above.

```
Inventory> db.products.find({ price: { $gte: 900 } })
{
  _id: ObjectId('6ab7df10fa70e9ba4d075df9'),
  productId: 3,
  name: 'Bluetooth Speaker',
  category: 'Electronics',
  price: 920,
  warehouse: 'Kanpur'
}
{
  _id: ObjectId('6ab7df10fa70e9ba4d075dfe'),
  productId: 8,
  name: 'Laptop Stand',
  category: 'Electronics',
  price: 900,
  warehouse: 'Lucknow'
}
{
  _id: ObjectId('6ab7df10fa70e9ba4d075e02'),
  productId: 12,
  name: 'USB Hub',
  category: 'Electronics',
  price: 950,
  warehouse: 'Lucknow'
}
{
  _id: ObjectId('6ab7df10fa70e9ba4d075e09'),
  productId: 19,
  name: 'Sofa Set',
  category: 'Furniture',
  price: 910,
  warehouse: 'Lucknow'
}
{
```

9. LESS THAN OR EQUAL TO – \$lte

MongoDB Query:

```
db.products.find({ price: { $lte: 700 } })
```

Explanation: Finds products whose price is 700 or below.

```

Inventory> db.products.find({ price: { $lte: 700 } })
{
  _id: ObjectId('6ab7df10fa70e9ba4d075dfa'),
  productId: 4,
  name: 'Sunflower Oil 1L',
  category: 'Grocery',
  price: 690,
  warehouse: 'Ayodhya'
}
{
  _id: ObjectId('6ab7df10fa70e9ba4d075e00'),
  productId: 10,
  name: 'Wheat Flour 5kg',
  category: 'Grocery',
  price: 650,
  warehouse: 'Kanpur'
}
{
  _id: ObjectId('6ab7df10fa70e9ba4d075e08'),
  productId: 18,
  name: 'Masala Tea',
  category: 'Grocery',
  price: 580,
  warehouse: 'Gonda'
}
{
  _id: ObjectId('6ab7df10fa70e9ba4d075e0a'),
  productId: 20,
  name: 'Wireless Earbuds',
  category: 'Electronics',
  price: 630,
  warehouse: 'Bareilly'
}
{

```

10. UPDATE OPERATION

MongoDB Query:

```

db.products.updateOne(
  { productId: 5 },
  { $set: { price: 950 } }
)

```

Explanation: Update the price to 950 of the product with productId 5.

```
Inventory> db.products.updateOne(
...   { productId: 5 },
...   { $set: { price: 950 } }
... )
... {
...   acknowledged: true,
...   matchedCount: 1,
...   modifiedCount: 1
... }
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
Inventory> █
```

11. DELETEONEOPERATION

MongoDB Query:

```
db.products.deleteOne({ productId: 10 })
```

Explanation: Deletes one document matching productId 10.

```
Inventory> db.products.deleteOne({ productId: 10 })
{ acknowledged: true, deletedCount: 1 }
Inventory> █
```

12. DELETEMANY OPERATION

MongoDB Query:

```
db.products.deleteMany({
  price: { $lt: 600 }
})
```

Explanation: Deletes all products having price below 600.

```
Inventory> db.products.deleteMany({
...   price: { $lt: 600 }
... })
... { acknowledged: true, deletedCount: 3 }
{ acknowledged: true, deletedCount: 3 }
Inventory> █
```

13. AND OPERATION—\$and

MongoDB Query:

```
db.products.find({
  $and: [
    { price: { $gt: 800 } },
    { category: "Electronics" }
  ]
})
```

```
]
})
```

Explanation: Finds products satisfying both conditions.

```
Inventory> db.products.find({
...   $and: [
...     { price: { $gt: 800 } },
...     { category: "Electronics" }
...   ]
... })
{
  _id: ObjectId('6ab7df10fa70e9ba4d075df7'),
  productId: 1,
  name: 'Wireless Mouse',
  category: 'Electronics',
  price: 880,
  warehouse: 'Lucknow'
}
{
  _id: ObjectId('6ab7df10fa70e9ba4d075df9'),
  productId: 3,
  name: 'Bluetooth Speaker',
  category: 'Electronics',
  price: 920,
  warehouse: 'Kanpur'
}
{
  _id: ObjectId('6ab7df10fa70e9ba4d075dfb'),
  productId: 5,
  name: 'LED Monitor',
  category: 'Electronics',
  price: 950,
  warehouse: 'Varanasi'
}
{
  _id: ObjectId('6ab7df10fa70e9ba4d075dfe'),
  productId: 8,
  name: 'Laptop Stand',
```

14. OR OPERATION—\$or

MongoDB Query:

```
db.products.find({
  $or: [
    { warehouse: "Lucknow" },
    { warehouse: "Kanpur" }
  ]
})
```

Explanation: Finds products stored in the Lucknow or Kanpur warehouse.

```

Inventory> db.products.find({
...   $or: [
...     { warehouse: "Lucknow" },
...     { warehouse: "Kanpur" }
...   ]
... })
{
  _id: ObjectId('6ab7df10fa70e9ba4d075df7'),
  productId: 1,
  name: 'Wireless Mouse',
  category: 'Electronics',
  price: 880,
  warehouse: 'Lucknow'
}
{
  _id: ObjectId('6ab7df10fa70e9ba4d075df9'),
  productId: 3,
  name: 'Bluetooth Speaker',
  category: 'Electronics',
  price: 920,
  warehouse: 'Kanpur'
}
{
  _id: ObjectId('6ab7df10fa70e9ba4d075dfc'),
  productId: 6,
  name: 'Office Chair',
  category: 'Furniture',
  price: 850,
  warehouse: 'Lucknow'
}
{
  _id: ObjectId('6ab7df10fa70e9ba4d075dfe'),
  productId: 8,
  name: 'Laptop Stand',

```

15. SORT OPERATION

A. MongoDB Query:

```
db.products.find().sort({ price: 1 })
```

Explanation: Sorts products by price from lowest to highest. Use 1 for ascending order.

```
Inventory> db.products.find().sort({ price: 1 })
{
  _id: ObjectId('6ab7df10fa70e9ba4d075e12'),
  productId: 28,
  name: 'Almonds Pack',
  category: 'Grocery',
  price: 610,
  warehouse: 'Gorakhpur'
}
{
  _id: ObjectId('6ab7df10fa70e9ba4d075e21'),
  productId: 43,
  name: 'Smart Bulb',
  category: 'Electronics',
  price: 620,
  warehouse: 'Ayodhya'
}
{
  _id: ObjectId('6ab7df10fa70e9ba4d075e0a'),
  productId: 20,
  name: 'Wireless Earbuds',
  category: 'Electronics',
  price: 630,
  warehouse: 'Bareilly'
}
{
  _id: ObjectId('6ab7df10fa70e9ba4d075e27'),
  productId: 49,
  name: 'Portable SSD',
  category: 'Electronics',
  price: 640,
  warehouse: 'Bareilly'
}
{
```

B. MongoDB Query:

```
db.products.find().sort({ price: -1 })
```

Explanation: Sorts products by price from highest to lowest. Use -1 for descending order.

```
Inventory> db.products.find().sort({ price: -1 })
{
  _id: ObjectId('6ab7df10fa70e9ba4d075e19'),
  productId: 35,
  name: 'Cooling Fan',
  category: 'Electronics',
  price: 960,
  warehouse: 'Prayagraj'
}
{
  _id: ObjectId('6ab7df10fa70e9ba4d075dfb'),
  productId: 5,
  name: 'LED Monitor',
  category: 'Electronics',
  price: 950,
  warehouse: 'Varanasi'
}
{
  _id: ObjectId('6ab7df10fa70e9ba4d075e02'),
  productId: 12,
  name: 'USB Hub',
  category: 'Electronics',
  price: 950,
  warehouse: 'Lucknow'
}
{
  _id: ObjectId('6ab7df10fa70e9ba4d075e10'),
  productId: 26,
  name: 'Wardrobe',
  category: 'Furniture',
  price: 940,
  warehouse: 'Lucknow'
}
{
```

16. LIMIT OPERATION

MongoDB Query:

```
db.products.find().limit(5)
```

Explanation: Display only the first five documents.

```
Inventory> db.products.find().limit(5)
{
  _id: ObjectId('6ab7df10fa70e9ba4d075df7'),
  productId: 1,
  name: 'Wireless Mouse',
  category: 'Electronics',
  price: 880,
  warehouse: 'Lucknow'
}
{
  _id: ObjectId('6ab7df10fa70e9ba4d075df8'),
  productId: 2,
  name: 'Basmati Rice 5kg',
  category: 'Grocery',
  price: 760,
  warehouse: 'Jaunpur'
}
{
  _id: ObjectId('6ab7df10fa70e9ba4d075df9'),
  productId: 3,
  name: 'Bluetooth Speaker',
  category: 'Electronics',
  price: 920,
  warehouse: 'Kanpur'
}
{
  _id: ObjectId('6ab7df10fa70e9ba4d075dfa'),
  productId: 4,
  name: 'Sunflower Oil 1L',
  category: 'Grocery',
  price: 690,
  warehouse: 'Ayodhya'
}
{
```

17. SEARCHBY WAREHOUSE

MongoDB Query:

```
db.products.find({ warehouse: "Lucknow" })
```

Explanation: Finds all products stored in the Lucknow warehouse.

```

Inventory> db.products.find({ warehouse: "Lucknow" })
{
  _id: ObjectId('6ab7df10fa70e9ba4d075df7'),
  productId: 1,
  name: 'Wireless Mouse',
  category: 'Electronics',
  price: 880,
  warehouse: 'Lucknow'
}
{
  _id: ObjectId('6ab7df10fa70e9ba4d075dfc'),
  productId: 6,
  name: 'Office Chair',
  category: 'Furniture',
  price: 850,
  warehouse: 'Lucknow'
}
{
  _id: ObjectId('6ab7df10fa70e9ba4d075dfe'),
  productId: 8,
  name: 'Laptop Stand',
  category: 'Electronics',
  price: 900,
  warehouse: 'Lucknow'
}
{
  _id: ObjectId('6ab7df10fa70e9ba4d075e02'),
  productId: 12,
  name: 'USB Hub',
  category: 'Electronics',
  price: 950,
  warehouse: 'Lucknow'
}
{

```

18. COUNT DOCUMENTS

MongoDB Query:

```
db.products.countDocuments()
```

Explanation: Counts documents currently present in the collection (performed after the delete operations).

```

Inventory> db.products.countDocuments()
46
Inventory> █

```

MONGODB OPERATORS USED:

Ope r ator	Meani ng	Ex ampl e
\$eq	Equal to	price: { \$eq: 880 }
\$gt	Greater than	price: { \$gt: 850 }
\$lt	Less than	price: { \$lt: 600 }
\$gte	Greater than or equal to	price: { \$gte: 900 }
\$lte	Less than or equal to	price: { \$lte: 700 }
\$and	All conditions true	\$and: [...]
\$or	At least one condition true	\$or: [...]

CONCLUSION

This project provided practical knowledge of NoSQL databases using MongoDB. A Product Inventory Management System containing product information was created and different MongoDB operations were performed on the data.

Through this project, I learned how to insert, retrieve, update, and delete documents. I also learned how to sort data in ascending and descending order, how comparison operators such as \$eq, \$gt, \$lt, \$gte, and \$lte work, along with logical operators such as \$and and \$or.

Overall, this project helped me understand how MongoDB can be used to manage and query product inventory data efficiently.